

TENANT GUIDE

HospitalOS

A complete guide to running your hospital on HospitalOS

For hospital administrators, clinicians, and staff

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1 Welcome to HospitalOS

HospitalOS is a full hospital operating system built for tertiary and teaching hospitals — the range of departments, workflows, and scale a facility like University College Hospital, Ibadan actually runs, not a simplified demo of one department.

How the system is organised

The sidebar is arranged by workflow, in roughly the order work happens through a hospital, not by org chart. Ten groups:

Group	What lives there
Overview	Dashboard, alerts, worklist, communication hub
Patient care	Registration/ADT, wards, ICU, emergency, maternity, records, renal, bookings
Diagnostics	Laboratory, blood bank, imaging (X-ray/US/CT/MRI), radiotherapy, instruments gateway
Pharmacy	Dispensing, inventory, formulary
Finance & trade	Billing, payments, claims, procurement, stores
Operations	Scheduling, CSSD, biomedical, facility, fleet, support, visitors
Academic	Training, logbooks, CME, research, ethics committee
Public health	Surveillance, immunisation, outreach, reporting
Intelligence	Analytics, reports, forecasting
Administration	Users & roles, facilities, instruments gateway, pricing, documents, audit, settings

Note. You only see the groups your role grants. A pharmacist's sidebar shows Overview and Pharmacy; a cashier's shows Overview and Finance. This keeps each person's view to what they actually use — it is not secrecy.

2 Signing in and understanding your role

Sign in with your hospital email. Your role, set by an administrator under **Administration** → **Users & roles**, determines two separate things:

- **Which areas you can open** — the sidebar groups and routes you can reach at all.
- **What you may do within an area** — the specific buttons available to you once there.

This two-level design enforces real separation of duties, not just visibility:

Role	Can	Cannot
Nurse	Admit patients, record vitals, file clinical notes	Discharge a patient
Lab Scientist	Collect samples, enter results, verify results	Reach Patient care at all
Radiographer	Order and report imaging studies	Verify laboratory results
Pharmacist	Dispense medication, restock inventory	Reach Diagnostics at all



Role	Can	Cannot
Cashier	Take payments, file insurance claims	Approve a claim
Records Officer	Register new patients	Admit or discharge a patient
Doctor	Full patient-care and diagnostics actions	Administration functions
Super Admin	Everything	—

Note. The last active Super Admin account cannot be deactivated by anyone, including another Super Admin — this prevents a hospital from accidentally locking itself out of its own system.

Important. The current sign-in check runs in the browser. It is the correct shape for a real authentication layer — every screen and button already defers to it correctly — but it does not yet verify credentials against a server. Treat this build as a controlled preview, not a production login.

3 Overview and the alert system

The Dashboard opens on live occupancy, admissions, and revenue charts. Every stat card links straight into the module behind it.

Alerts — eleven sources, one feed

Overview → Alerts is a single feed for the entire hospital. The red badge on the sidebar refreshes every few seconds.

Source	Raises an alert when...	Severity
Laboratory	A result crosses a panic threshold	Critical
Critical care	Any ICU/HDU vital goes critical	Critical
Radiology	A report is flagged with an urgent finding	Critical
Maternity	A newborn's Apgar score is below 7	Critical
Blood bank	A group is below reorder / a unit nears expiry	Critical / Warning
Pharmacy	A drug is out of stock / below reorder	Critical / Warning
Instruments	A device errors or goes offline	Critical / Warning
Renal	A scheduled dialysis session is overdue	Critical
Operations	Equipment under repair / a vehicle is out of service	Warning
Oncology	A chemotherapy cycle is overdue	Warning
Public health	A notifiable disease is trending upward	Warning

Most alerts do not need to be acknowledged — they read the live state, so fixing the underlying problem (restocking a drug, stabilising a patient) clears the alert automatically. Acknowledging simply moves an alert out of the active feed for record-keeping.



Notifications

The bell icon in the topbar currently surfaces booking requests awaiting review. Its count does not shrink on its own — a booking only leaves the bell once someone actually confirms or declines it. Nothing here times out or dismisses itself.



4 Patient care

Registration, admission, and the ward board

Registration & ADT is where a patient's journey begins — register, then admit to a bed. The bed board and ADT read the same registry, so a bed taken in one is unavailable in the other instantly; double-booking is not possible.

Accommodation tiers

Seven accommodation tiers, each with its own nightly rate, feeding automatically into billing:

Tier	Rate / night
General Ward	■15,000
Semi-Private	■35,000
Private Room	■60,000
Private Suite	■120,000
VIP Suite	■220,000
Executive Suite	■350,000
Critical Care	■180,000

Note. These are catalogue defaults — your hospital can set its own rates for every tier. See Section 11.

The medical record

Medical records holds the clinical picture per patient: SOAP-structured notes, an ICD-10 problem list, and recorded allergies. **Filed notes cannot be edited or deleted** — a correction is filed as a new amendment that references the original, with both remaining visible. This mirrors how paper charting actually works.

Important. A severe allergy on a patient's record blocks Pharmacy from dispensing that drug outright — the dispense button is disabled with no override in the interface.

ICU/HDU, Emergency, Maternity, Renal & Dialysis, Online Bookings

- **ICU/HDU** tracks five vitals per bed with panic thresholds; any critical reading marks the patient Unstable and raises an alert automatically.
- **Emergency** sorts by triage acuity first, arrival time second — a Level 1 arrival outranks a Level 4 who has waited longer.
- **Maternity** tracks labour stage through to delivery; a newborn Apgar below 7 raises a critical alert immediately.
- **Renal & dialysis** runs a haemodialysis programme (vascular access, schedule, per-session fluid removal) and a CKD registry that auto-stages from eGFR.
- **Online bookings** holds appointment requests from your website. Confirming and checking in a booking calls the same function Outpatient uses internally — a checked-in booking is a genuine addition to today's clinic queue, not a separate list.

5 Diagnostics



41 laboratory tests across eight disciplines (Chemistry, Haematology, Microbiology, Serology, Endocrinology, Molecular, Histopathology, Transfusion Medicine), and 27 imaging protocols across Ultrasound, CT, MRI, General Radiography, and Mammography.

Laboratory

Orders move Ordered → Collected → Resulted → Verified — you cannot skip a stage. Each result flags Low/High/Critical live against its reference range; a critical value raises an alert the instant it is saved.

Ultrasound, CT and MRI

Three modality-specific worklists, each a filtered view of the exact same study records the general Radiology screen uses — requesting a study anywhere shows it everywhere instantly. Each modality captures parameters appropriate to it: Ultrasound records probe and Doppler use, CT records contrast and slice thickness, MRI records sequence protocol and field strength.

Instruments & devices gateway

One interoperability hub for everything your hospital connects, reachable from both Diagnostics and Administration:

Category	Protocol	Examples
Laboratory analyzer	HL7 v2 / MLLP	Sysmex, Cobas, a 2009 Beckman AU480 still in service
Imaging modality	DICOM 3.0	CT, MRI, ultrasound, a 2007 CR reader
Radiotherapy system	DICOM-RT / proprietary	A modern LINAC and a manual Cobalt-60 unit
Printer	IPP / raw socket / serial	Label printers, report printers, a legacy dot-matrix printer

Old and modern equipment sit side by side deliberately — a real hospital's fleet spans decades of purchase dates.

Biobanking & lab utilities

Biobanking tracks long-term specimen storage across four capacity-limited units, separate from the active testing worklist. Lab utilities holds seven clinical calculators (eGFR, creatinine clearance, BMI, BSA, fluids, anion gap, corrected calcium), unit converters, and bench reference material.



6 Pharmacy

Dispensing checks the patient's recorded allergies before completing a transaction, blocks any quantity greater than available stock, and computes the naira total live. Inventory alerts clear themselves the moment stock is restocked above the reorder level.

7 Finance & trade

Billing aggregates charges from five sources automatically — nobody re-keys a figure: laboratory, pharmacy, radiology, theatre procedures, and accommodation bed-nights. A theatre case becomes billable the moment it enters theatre, not when scheduled; a radiology study, the moment it is performed, not requested.

Claims move Submitted → Approved/Rejected → Paid. As shown in Section 2, a cashier can file a claim but cannot approve it — separation of duties, not an oversight.

8 Operations, Academic & Public health

Operations covers shift scheduling, CSSD sterilisation cycles, biomedical equipment, facility status, fleet, support services, and visitor management.

Ethics committee (IRB)

A real submission-to-decision workflow: Submitted → Under review → one of Revisions requested / Approved / Rejected. Every decision requires a written reviewer comment — the decision button stays disabled without one — and a finalised decision cannot be silently reopened.

Public health covers disease surveillance, immunisation coverage, community outreach, and IDSR/NHMIS-style reporting.

9 Intelligence

Analytics, printable operational reports, and 7-day occupancy/revenue forecasting. Diagnostic intelligence (under Diagnostics) reads across Laboratory, Radiology, and Blood Bank to surface test utilisation, department load, and declared-versus-actual turnaround time.



10 Administration

Screen	What it does
Users & roles	Create staff accounts, assign roles, view the area/action permission matrix
Facilities & sites	Manage multi-site hospital records
Instruments gateway	The device hub described in Section 5
Pricing	Set your own prices — see Section 11
Documents & templates	Upload and manage your own files — see Section 11
Security & audit	The hash-chained audit trail — see Section 11
Settings	Hospital name, logo, currency, timezone, toggles

11 Making HospitalOS your own

Setting your own prices

Administration → **Pricing** lets you override the catalogue default for any test, drug, imaging study, procedure, or accommodation tier. Every billing calculation and every price shown to staff reads through the same value — changing a price here changes what is actually charged, for every new transaction from that moment on.

Note. Past charges are never rewritten. A patient billed before you changed a price keeps their original total; only new transactions use the new price.

Your branding

Administration → **Settings** has a hospital name field and a logo URL field. Both appear top-right on every screen, live, within seconds of saving — distinct from the HospitalOS product branding in the sidebar. No logo set shows your initials instead.

Your documents

Administration → **Documents & templates** accepts real uploads — drag a file in or browse, assign any category you like (there is no fixed list; type a new one whenever you need it), and it is immediately downloadable, renameable, and deletable.

Important. Uploaded files live in your browser session only until server-side storage is connected — they do not survive a page reload. Keep your own copy of anything important in the meantime.

The audit trail

Administration → **Security & audit** is append-only by construction: entries are frozen the moment they are written, and there is no update or delete API at all. Each entry is hash-chained to the one before it, so editing a past record breaks verification at that record, and deleting one breaks verification at the next.

Note. This is *tamper-evident*, not *tamper-proof* — a browser cannot stop someone with developer tools. Its value is that the same design carries over unchanged to server-side storage, where it becomes a genuine control.



12 Getting help

Global search

Press **Ctrl+K** (Cmd+K on Mac) from anywhere. It searches screens your role can reach, patients by name or hospital number, and the full help library, all at once.

Help & documentation

Pinned at the bottom of the sidebar, outside the scrollable workflow groups, reflecting that it is a standalone reference — not tucked inside any one department. Every article covers what a screen does and, just as importantly, why it works the way it does.

13 What to know before you rely on this build

This is an honest account, not boilerplate. Read it before entering anything you cannot afford to lose.

- **Data does not persist.** Every module runs on an in-memory data layer for this preview. Records reset when the page reloads. Each service is deliberately written as an async API shaped like a real network call, so a database swap does not require rebuilding any screen.
- **Sign-in is not yet secure.** Credentials are checked in the browser. The structure is correct — every guard already defers to the session it sets — but it protects nothing until a server verifies credentials.
- **The audit trail detects tampering; it does not prevent it client-side.** See Section 11.
- **The instruments gateway simulates live listening.** Device management and monitoring are real; opening an actual network socket (MLLP, DICOM, a print daemon) is server territory a browser cannot reach into.

Important. Do not enter real patient data into this build. It is a working preview of a production system, not production itself.